Features

- It allows users to create a neural network model without programming
- Offers a good user interface in terms of data management and result interpretation using the neural designer tool
- Provides faster training compared to other deep learning frameworks
- Outstanding performance in terms of execution speed and memory allocation

5. Microsoft Cognitive Toolkit (CNTK)

Microsoft Cognitive Toolkit (CNTK) is a commercial grade open source deep learning framework. Its primary release was on January 25, 2016. For more details, visit *docs.microsoft.com/en-us/cognitive-toolkit.*

Features

- CNTK APIs are available in Python, C# as well as C++. Also provides its own model programming language such as BrainScript
- Evaluation of the CNTK model feature is supported by Java Apps
- Provides low-level and high-level APIs
- Provides an easy and simple way to combine various deep learning models like a convolutional neural network, recurrent neural network, and deep feed-forward neural network
- Autonomous differentiation and parallelisation over several GPUs and servers
- Supports implementation of supervised learning, unsupervised learning, reinforcement learning, as well as generative adversarial networks

6. Apache MXNet

Apache MXNet *(mxnet.apache.org)* is an open source deep learning framework released (version 0.7.0) on May 26, 2016.

Table 2: A comparison of open source deep learning tools

Tool	Licensed by	Written in	CUDA support
TensorFlow	Apache 2.0	C++, Python, CUDA	Yes
Keras	MIT	Python	Yes
PyTorch	BSD	Python, C, C++, CUDA	Yes
OpenNN	GNU LGPL	C++	Yes
CNTK	MIT	C++	Yes
MXNet	Apache 2.0	Small C++ core library	Yes
Deeplearning-Kit	Apache 2.0	Metal, Swift	No
Deeplearning4J	Apache 2.0	C++, Java	Yes
Darknet	YOLO	C, CUDA	Yes
PlaidML	Apache 2.0	Python, C++, OpenCL	No

Features

- Provides hybrid front-end with seamless transition between Gluon eager imperative mode and symbolic mode; high speed and flexibility
- Supports dual parameters server and Horovod (distributed deep learning framework) to offer scalable distributed training and performance optimisation
- Provides comprehensive and flexible Python APIs and supports other languages such as Scala, Julia, Clojure, Java, C++, R and Perl
- Supports computer vision, natural language processing, text classification, and timeseries

7. DeepLearningKit

DeepLearningKit is an open source deep learning framework developed for the Apple iOS, OS X, and tvOS. For more details, visit *github.com/DeepLearningKit/DeepLearningKit.*

Features

- Developed using Metal and Swift for GPU acceleration and app integration, respectively
- Uses convolutional neural networks for image recognition on Apple devices

8. Deeplearning4J

Deeplearning4J is an open source deep learning framework that was primarily released in 2014, with a stable release on May 13, 2020. The main authors are Alex D. Black, Adam Gibson, Vyacheslav Kokorin, and Josh Patterson. For more details, visit *deeplearning4j.konduit.ai.*

Features

- Supports all the Java virtual machine based languages such as Java, Scala, Clojure, and Kotlin
- Provides a distributed deep learning framework by felicitating training in a cluster. The framework is supported by distributed systems Apache Spark and Hadoop
- Has a designed vector space model and topic model to manage large size text sets and perform natural